

Please write clearly in block capitals.

Centre number

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

Candidate number

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

Surname

---

Forename(s)

---

Candidate signature

---

# AS BIOLOGY

## Paper 2

Friday 24 May 2019

Morning

Time allowed: 1 hour 30 minutes

### Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

### Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Show all your working.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for the questions are shown in brackets.
- The maximum mark for this paper is 75.

#### For Examiner's Use

| Question     | Mark |
|--------------|------|
| 1            |      |
| 2            |      |
| 3            |      |
| 4            |      |
| 5            |      |
| 6            |      |
| 7            |      |
| 8            |      |
| 9            |      |
| <b>TOTAL</b> |      |



J U N 1 9 7 4 0 1 2 0 1

**There are no questions on this page**

*Do not write  
outside the  
box*

**DO NOT WRITE ON THIS PAGE  
ANSWER IN THE SPACES PROVIDED**



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | . | 1 |
|---|---|---|---|

**[4 marks]**

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

---

---

---

---

**Turn over ►**



Scientists investigated how the diet of rabbits affected their digestion and absorption of protein. The scientists fed rabbits an identical mass of food but varied the percentage of protein in the food.

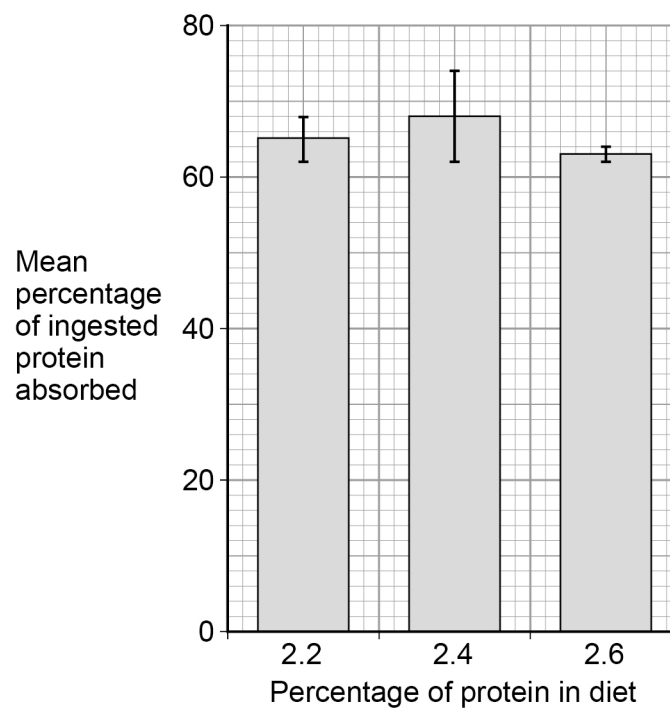
The scientists measured the mean mass of protein fed to the rabbits that was absorbed, which they then expressed as a percentage value.

The scientists' results are shown in **Figure 1**.

The error bars show  $\pm 2$  standard deviations.

$\pm 2$  standard deviations cover 95% of the data.

**Figure 1**



What can you conclude about the absorption of the products of protein digestion as the percentage of protein increased in the rabbits' food?

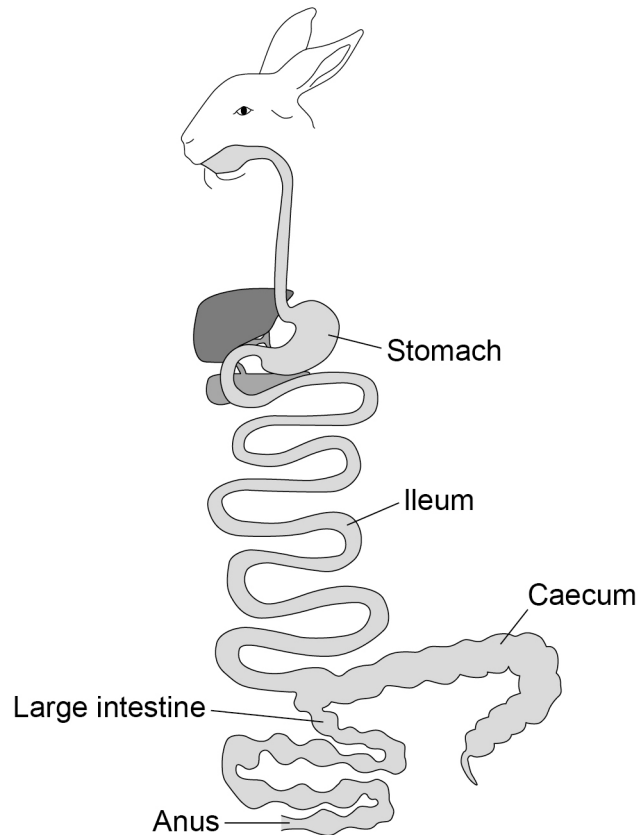
[illegible]

**Turn over ►**



The digestive system of a rabbit is shown in **Figure 2**.

**Figure 2**



**0 1 . 3**

The food eaten by a rabbit is digested mainly by microorganisms in its caecum. The caecum is a section of intestine attached between the ileum and the large intestine. The resulting semi-digested material leaves the anus of a rabbit as soft, caecal droppings. The rabbit then eats these caecal droppings.

Use this information and **Figure 2** to suggest how eating its own caecal droppings helps a rabbit's digestion and absorption of dietary protein.

**[3 marks]**

---

---

---

---

---

---

---



10

**Turn over ►**



0 2 . 1

Describe and explain the effect of increasing carbon dioxide concentration on the dissociation of oxyhaemoglobin.

[2 marks]

---

---

---

---

---

---

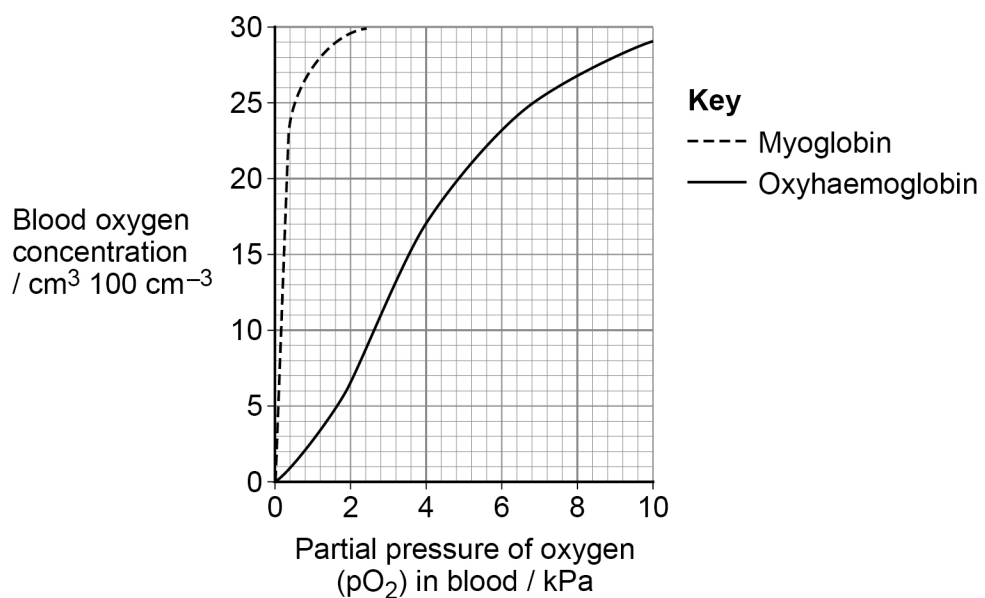
---

---

Seals are diving mammals. They fill their lungs with air before they dive and hold their breath during the dive.

**Figure 3** shows the dissociation curves for seal oxyhaemoglobin and seal myoglobin. Myoglobin is an oxygen-carrying protein found in muscles.

**Figure 3**



0 2 . 2

Use information in **Figure 3** to explain how the seal's myoglobin dissociation curve shows the seal is adapted for diving.

[2 marks]

---

---

---

---

---

---

---

---

[Extra space]

---

---

0 2 . 3

Scientists measured the oxygen carrying capacity of seal blood. They found the haemoglobin in a 190 kg seal contained  $1.07 \times 10^4 \text{ cm}^3$  oxygen. When the seal dived, it used  $5.2 \text{ cm}^3$  oxygen per minute per kg of body mass.

Use this information to calculate the maximum number of minutes the seal can remain under water. Assume that all of the oxygen attached to the haemoglobin is released during the dive.

[2 marks]

Answer = \_\_\_\_\_ minutes

6

Turn over ►



**0 3 . 1**

Name the **two** scientists who proposed models of the chemical structure of DNA and of DNA replication.

**[1 mark]**

A scientist replicated DNA in a test tube. To do this, he mixed an enzyme with identical single-stranded DNA fragments and a solution containing DNA nucleotides.

**0 3 . 2**

Name the enzyme used in this DNA replication.

**[1 mark]****0 3 . 3**

Use your knowledge of semi-conservative replication of DNA to suggest:

**[3 marks]**

1. the role of the single-stranded DNA fragments

---

---

---

---

2. the role of the DNA nucleotides.

---

---

---

---



|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 1 |
|---|---|---|---|

Describe and explain the role of antibodies in stimulating phagocytosis.

Do **not** include details about the process of phagocytosis.

**[2 marks]**

---

---

---

---

---

---

---

---

**Question 4 continues on the next page**

**Turn over ►**



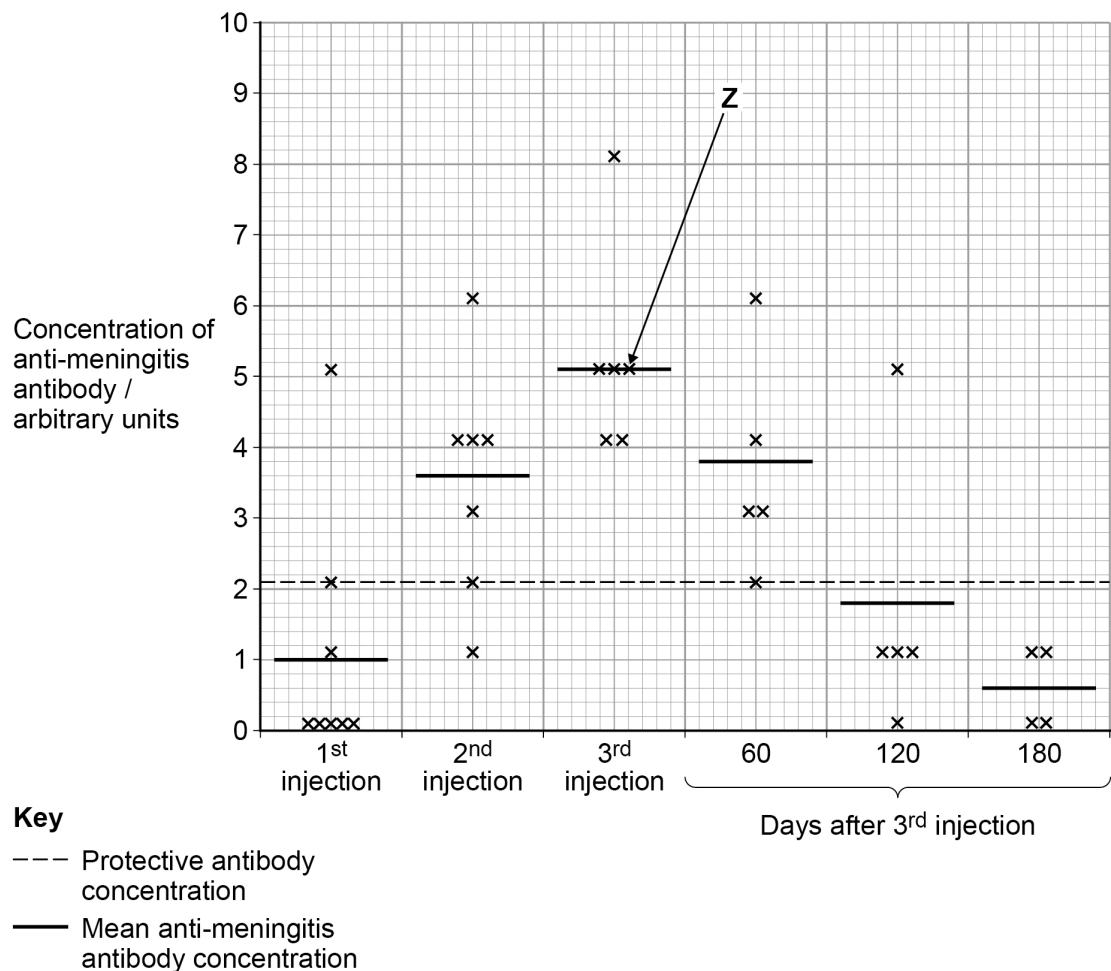
*Meningococcus* bacteria cause a disease called meningitis. Scientists investigated a new meningitis vaccine (MenG) by measuring changes in blood anti-meningitis antibody concentration in mice.

Each mouse was given three separate MenG injections. The concentration of anti-meningitis antibody was measured in a sample of blood taken soon after each injection.

After the 3rd injection, the concentration of anti-meningitis antibody in the blood was also measured after 60 days, after 120 days and then after 180 days.

**Figure 4** shows the scientists' results. Each plotted point in **Figure 4** is the result for a different mouse.

**Figure 4**



Do not write  
outside the  
box

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 2 |
|---|---|---|---|

The scientists discovered that the concentration of anti-meningitis antibody of the mouse labelled **Z** in **Figure 4** decreased after the 3rd injection at a constant rate of 0.027 arbitrary units per day.

Use this information and **Figure 4** to calculate the number of days after the 3rd injection the antibody concentration is higher than the protective antibody concentration for this mouse.

**[2 marks]**

Answer =                      days

|   |   |   |   |
|---|---|---|---|
| 0 | 4 | . | 3 |
|---|---|---|---|

Using **Figure 4**, what can you conclude about the effectiveness of each injection on the immune response of these mice?

**[4 marks]**

[illegible]

**Turn over ►**



0 4 . 4

The scientists hypothesised that memory B cells had formed in the mice 180 days after the 3rd injection.

Suggest and explain a practical method the scientists could use to test this hypothesis.

[2 marks]

---

---

---

---

---

---

---

---

---

---

[Extra space]

---

---

---

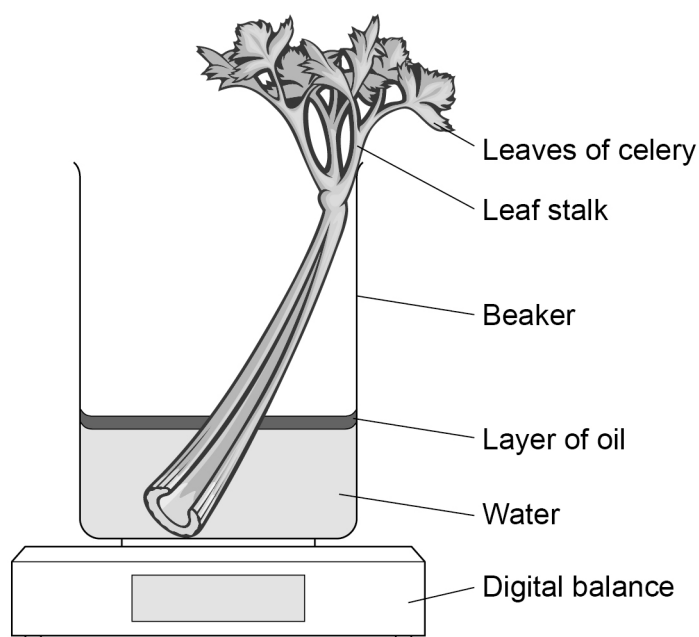
10



0 5

A student used the apparatus shown in **Figure 5** and a digital balance to determine the rate of water movement in a celery stalk in grams per hour per group of xylem vessels.

**Figure 5**



0 5 . 1

The student measured the time taken for water movement.  
Give **two** other measurements he made to calculate the rate of water movement.

**[2 marks]**

1 \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

2 \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

0 5 . 2

Give the reason for adding a layer of oil to the water in the beaker.

**[1 mark]**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Question 5 continues on the next page**

**Turn over ►**



A different student used coloured water to investigate the movement of water in leaf stalks of celery.

- cut equal lengths of stalk from each plant
- put the cut end of each stalk into coloured water
- left these stalks to take up the coloured water for 20 minutes
- used a sharp scalpel to cut slices from the stalks at 1 mm intervals until she reached a slice with no coloured water.

**Figure 6** shows a slice of leaf stalk with coloured water inside groups of xylem vessels.

Coloured water in a group of xylem vessels

**[3 marks]**

[illegible]

0 5 . 4

The student used a sharp scalpel to cut the celery. Describe how she should ensure she handled the scalpel safely during this procedure.

**[2 marks]**


---



---



---



---



---

The student measured the distance the coloured water had travelled in eight celery stalks.  
Her results are shown in **Table 1**.

**Table 1**

| Distance / mm |    |    |    |    |    |    |    |
|---------------|----|----|----|----|----|----|----|
| 70            | 35 | 40 | 35 | 30 | 80 | 42 | 44 |

0 5 . 5

The student had to choose whether to summarise her measurements by calculating the mean, the median or the mode.

Circle the most appropriate measure for this set of measurements.  
Give a reason for your choice and find the value using the measurements from all eight stalks.

**[2 marks]**

Mean\*

Median\*

Mode\*

\*circle one word.

Reason:

---



---

Calculation:

Answer = \_\_\_\_\_

10

**Turn over ►**

**0 6**

A scientist crossed a strain of the fungus *Neurospora* producing pink spores with a strain of *Neurospora* producing white spores.

To cross these strains, he used aseptic techniques. He moved a small agar cube containing one strain of the fungus onto a new agar plate. Then he placed a second agar cube containing the other strain of fungus next to the first agar cube.

**0 6****1**

Describe and explain **three** ways in which the scientist would ensure he used aseptic techniques to move each cube of agar onto a new agar plate.

**[3 marks]**

1

---



---

2

---



---

3

---



---

In the life cycle of *Neurospora* most stages are haploid. Fusion of two haploid strains of this fungus produces diploid zygotes. Nuclear division in these zygotes occurs by meiosis.

**0 6****2**

Give **two** differences between mitosis and meiosis.

**[2 marks]**

1

---



---

2

---



---

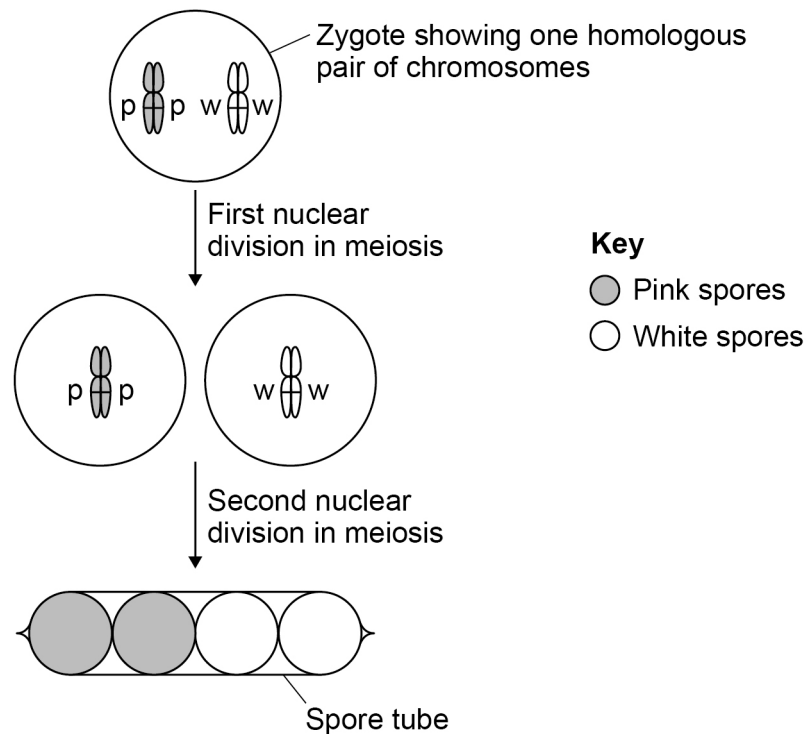


At the end of meiosis, this fungus produces cells called spores. The spores are produced in narrow tubes that restrict their movement. As a result, each tube contains a single line of spores. The spores are coloured either pink or white.

The spore colour gene is located on a pair of homologous chromosomes. Each zygote produced in this cross has one chromosome with a pink allele (p) and one chromosome with a white allele (w).

This is shown in **Figure 7**.

**Figure 7**



0 6 . 3

There are seven chromosomes in a spore nucleus.

Place a tick (✓) in the box next to the number that represents the number of **chromatids** present in the zygote shown in **Figure 7**.

[1 mark]

7

☐

14

☐

21

☐

28

☐

Question 6 continues on the next page

Turn over ►



## Table 2

| Type of spore tube | Arrangement of coloured spores                                                    | Number of narrow tubes |
|--------------------|-----------------------------------------------------------------------------------|------------------------|
| 1                  |  | 81                     |
| 2                  |  | 78                     |
| 3                  |  | 10                     |

Using all the information in this question, what can you conclude from the scientist's results about the movement of chromosomes in meiosis in this fungus?

**[3 marks]**

[illegible]

07

Scientists investigated changes in the mass of fish from three populations of the same species. The fish they used had a life cycle of one year.

The scientists set up three fish tanks, each containing a separate population. Each year the scientists removed all the fish from each tank and determined the mean mass of the fish removed. They then put back 10% of each population in the following way.

Tank **A** – put back only the largest fish.

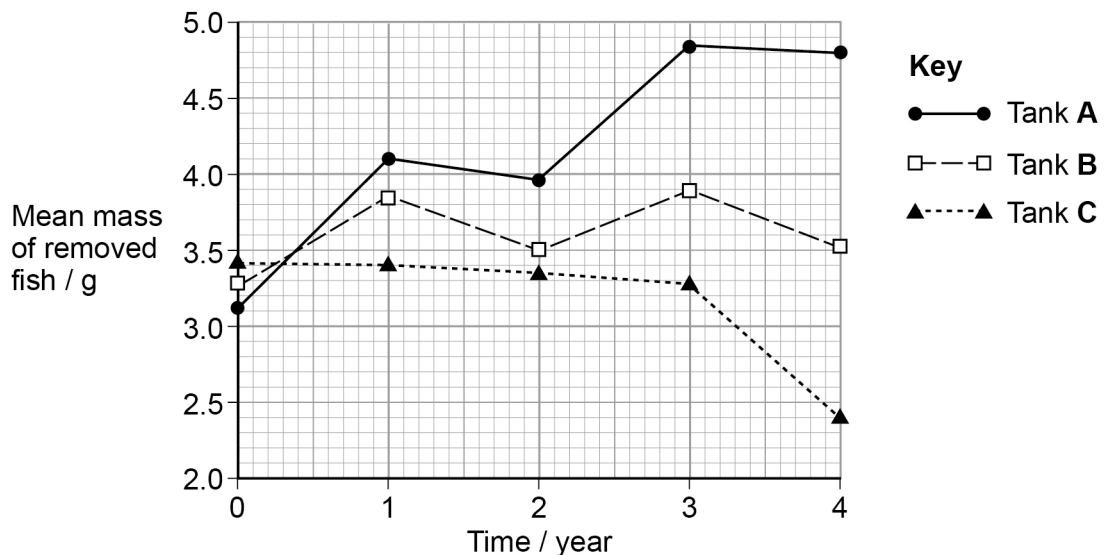
Tank **B** – put back fish at random.

Tank **C** – put back only the smallest fish.

During each year the fish were left to grow and reproduce.

The scientists' results are shown in **Figure 8**.

**Figure 8**



07.1

What type of selection were the scientists modelling in this investigation by putting back only the largest or only the smallest fish in Tank **A** and Tank **C**? Give a reason why.

**[2 marks]**

Type of selection \_\_\_\_\_

Reason \_\_\_\_\_

---



---



---

**Question 7 continues on the next page**

**Turn over ►**



**0 7 . 2** Explain the purpose of Tank **B** (on page 21).

**[2 marks]**

---

---

---

---

---

---

---

---

**0 7 . 3** Calculate the ratio of the mean mass of fish removed from Tank **A** (on page 21) to the mean mass of fish removed from Tank **C** (on page 21) at 1 year **and** at 4 years.

How much greater is the ratio at 4 years compared with the ratio at 1 year?

**[2 marks]**

Ratio at 1 year = \_\_\_\_\_

Ratio at 4 years = \_\_\_\_\_

How much greater at 4 years = \_\_\_\_\_



Sea fishing is regulated in law. The size of the mesh used in some fishing nets is controlled so that small fish can escape but large fish are captured. This regulation is designed to protect populations of wild fish.

Using all the information in this question, evaluate whether the scientists' investigation supports the use of these types of nets in sea fishing.

[illegible]

---

---

---

9

**Turn over for the next question**

**Turn over ►**



0 8 . 1

The genetic diversity of species is measured by comparing differences in the base sequence of DNA or differences in the base sequence of mRNA.

Give **two** other ways in which genetic diversity between species is measured.

[2 marks]

1

---



---



---

2

---



---



---

Scientists investigated differences between 260 North American bird species by comparing the base sequence of a gene in mitochondrial DNA. They compared the gene base sequence of each bird with all of the other 259 species. For each comparison they calculated the percentage difference in base sequence.

0 8 . 2

**Figure 9** shows the base sequence for part of the gene in two species.

**Figure 9**

Species 1    A   G   C   T   G   C   C   T   A   G   A

Species 2    A   T   G   T   G   G   C   A   A   G   A

Calculate the percentage difference in base sequence for these base sequences.

[1 mark]

Answer = \_\_\_\_\_ %

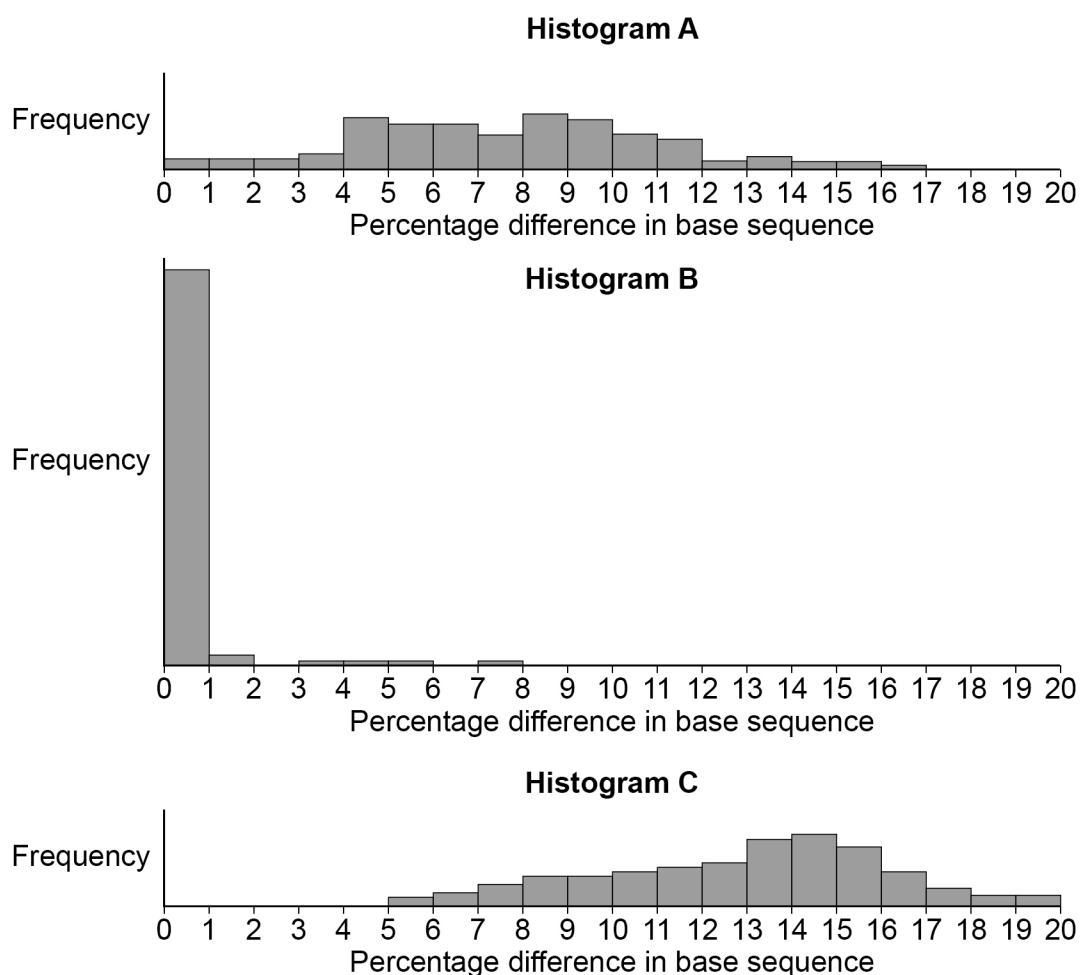


The scientists compared base sequences in:

- birds of the same species
- birds of different species in the same genus
- birds of different species in the same family.

The scientists' results are shown in **Figure 10**.

**Figure 10**



**0 8 . 3**

Complete **Table 3** by writing **A, B** or **C** in the box to correctly match the statement to each histogram shown in **Figure 10**.

**[1 mark]**

**Table 3**

| Statement                                    | Histogram |
|----------------------------------------------|-----------|
| Base sequences of birds of the same species. |           |
| Base sequences of birds of the same genus.   |           |
| Base sequences of birds of the same family.  |           |

**Question 8 continues on the next page**

**Turn over ►**



0 8 . 4

To calculate the percentage difference in base sequences, the scientists first counted the number of bases and the number of base differences.

What statistical test should the scientists use to test whether the number of base differences between birds in histogram **A** (on page 25) and birds in histogram **C** (on page 25) is statistically significant?

Place a tick (✓) in the box against the statistical test you would use.

Justify your answer.

**[2 marks]**

Chi-squared

☐

Correlation coefficient

☐

Student's t-test

☐

Justification

---



---



---



---

6



**[5 marks]**

[illegible]

---

---

---

---

**Turn over ►**



**[5 marks]**

[illegible]

**10**

### Copyright information

Copyright © 2019 AQA and its licensors. All rights reserved.

